

NICO MASTAKAS

Senior Product Engineer | Full-Stack TypeScript · React Native · Flutter | AI-Native Products: LLMs, Vision, RAG

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SUMMARY

Product Engineer & Full-Stack Software Engineer with 6+ years shipping production software end-to-end across mobile, web, and backend. I build full products as a one-person team, using Claude Code and Cursor as a force multiplier. Currently Senior Engineer at Input Output (IOHK); previously scaled a FinTech consumer app to 100K+ users at Bolster as one of four founding engineers, and co-founded a crypto-education startup. Outside of work I ship AI-native products end-to-end — LLM vision, RAG with hybrid retrieval, prompt-evaluation CI, and fine-tuned on-device models (see Selected Projects below).

EXPERIENCE

Senior Software Engineer — Input Output (IOHK) May 2024 – Present

Full-Stack TypeScript · Mobile · AI-Native Workflows United States · Remote

Lace — multi-chain crypto wallet (Cardano, Bitcoin, Midnight) shipped as browser extensions and native iOS/Android.

Working full-stack across mobile, extension, shared core, blockchain integrations, and the team's AI-augmented workflow.

- Architected a contract-module system (70 domain contracts, 43 modules) with runtime feature-flag loading via PostHog; modules import only via typed contracts. Designed a reactive state layer (Redux + RxJS) with selective persistence (AsyncStorage / SecureStore / MMKV) and encrypted storage.
- Shipped production code across React Native (Expo 52), browser extension (Webpack 5), and shared core in an Nx monorepo (~120 packages, 2,400+ TS files, 3,500+ commits). Integrated Cardano SDK (HD wallets, multi-sig, UTxO selection), Midnight Network, Bitcoin, and Ledger hardware wallets.
- Built a custom Claude Code framework for the team: 7 specialized agents, 4 commands, 4 MCP servers (Sequential Thinking, Context7, Atlassian, Figma). GitHub Actions + Claude Code for automated PR review via @claude mentions.
- Testing pyramid: 224 Vitest unit tests, 100+ Storybook 9 stories, WebdriverIO E2E on BrowserStack. CI/CD with EAS, Fastlane, Sentry source-maps.

Senior Software Engineer — Bolster Jan 2022 – May 2024

Full-Stack · React Native · Node.js · AWS Serverless United States · Remote

US FinTech credit-score app. Joined as one of 4 founding engineers; shipped greenfield from first commit, owning React Native frontend and Node.js serverless backend on AWS — scaled to 100K+ active users in 7 months.

- Reduced cold-start latency 40% (Hermes, bundle splitting, lazy-loading; +20% user satisfaction). Cut crash rate 35%, sustaining a crash-free rate above 98% across 50+ deployments. Led 3 major RN upgrades (0.63 → 0.71); test coverage 60% → 85%.
- Built a Node.js serverless backend on AWS from scratch — Lambda, API Gateway, DynamoDB, S3, Cognito, SNS/SQS — powering credit-score data and 3rd-party integrations (Plaid, credit bureaus). IaC with CloudFormation. Automated App Store / Play Store releases (Fastlane + GitHub Actions): 2h → 15min.

Senior Software Engineer — Globant Jan 2021 – Jan 2022

Full-Stack · React · React Native · Node.js | Client: Lessen (US FinTech / PropTech) Remote

Lessen — US fintech / property-tech platform on track to unicorn status. Shipped across the full stack: React web, React Native mobile, and Node.js / Hasura GraphQL backend, partnering directly with Lessen's product team.

- Architected a modular React Native component library (atomic-design, -30% cross-platform dev effort). Worked on the Node.js + Hasura GraphQL backend — schemas, resolvers, geospatial queries (PostgreSQL + PostGIS).
- Reduced production bugs 40% (Cypress E2E, Jest). Optimized CI/CD: 98% success rate, release cycles 3h → 20min. Mentored 4 junior engineers, +15% team velocity.

Co-Founder & Full-Stack Engineer — Moony Jan 2020 – Sep 2021

Founding Engineer · React · React Native · Node.js · GraphQL Argentina

- Co-founded a crypto-education startup. Built the product end-to-end — web (React), mobile (React Native), backend (Node.js + Express + GraphQL on PostgreSQL). Onboarded 500+ users in 3 months. Migrated REST → GraphQL (200ms → 120ms). Automated CI/CD (GitHub Actions, Docker): deploy 45min → 10min.
- Led technical hiring (team 2 → 6) and set engineering standards (TypeScript-first, atomic design, CI conventions) under early-stage uncertainty.

SELECTED PROJECTS (solo, self-directed)

Settle — AI invoice review for waste haulers (Claude Vision + human-in-the-loop)

Reads fuel/tipping invoices with Claude Vision and flags hidden markups against vendor history, gated by an approval queue. *Next.js · TypeScript · Claude · Neon · Drizzle*

Hiring Radar — AI job-matching agent (tool-use loop + long-term memory) with hybrid retrieval & an MCP server

Hybrid semantic + full-text search (recall@10 0.52 vs 0.28 semantic-only) on pgvector HNSW + RRF; autonomous agent with crash-recovery and cross-run memory; incremental Inngest pipeline; MCP server + prompt-injection defenses. *Next.js · TypeScript · pgvector · MCP · Inngest · Neon*

Shoebox — 100% on-device receipt digitizer — OCR + fine-tuned local model, benchmarked vs cloud

QLoRA fine-tune (MLX) reaching ~90% of Claude-Vision accuracy at \$0, fully offline (80% vs 72% base, published to Hugging Face Hub), plus a scikit-learn category classifier. *Python · PaddleOCR · OpenCV · MLX · Ollama · scikit-learn*

Rubric — CI for prompts — golden sets + LLM-as-judge

Calibrated LLM-as-judge (Cohen's κ 0.81) that fails the PR when a prompt or agent change regresses output quality. *TypeScript · Next.js · Turso · Groq · GitHub Actions*

Daily News — full-stack news app with an “Ask the News” RAG assistant

Answers grounded in the articles with tappable citations (local embeddings + Groq via a Cloud Function); Clean Architecture + BLoC, security rules tested in CI. *Flutter · Dart · Firebase · RAG*

Also: TodayOnly — native SwiftUI + SwiftData (macOS) and Tauri/Rust (Windows) · Tether WDK — non-custodial RN wallet with Secure-Enclave key storage · full portfolio at github.com/Qweffy.

SKILLS

Languages & Frontend: TypeScript, JavaScript, Python, Dart, Swift, React, React Native, Flutter, Next.js, Expo, SwiftUI, Redux Toolkit, RxJS, Tailwind/NativeWind

AI / LLM: Anthropic API (Claude, incl. Vision), RAG (hybrid search BM25 + vectors, RRF re-ranking, recall@k eval), pgvector / HNSW, vector embeddings, AI agents (tool-use loops + long-term memory), MCP servers (built + consumed), prompt engineering & injection defense, LLM-as-judge / evals, structured outputs, tool calling

ML & Data Science: Fine-tuning (QLoRA / MLX), quantization (GGUF), classical ML (scikit-learn: TF-IDF, cross-val, F1), OCR (PaddleOCR / OpenCV), local LLM serving (Ollama), Hugging Face Hub, pandas / EDA

Backend & Data: Node.js, Express, tRPC, GraphQL (Hasura), Firebase, PostgreSQL (Neon) + pgvector, Drizzle, Turso, MongoDB, Redis, SQLite, REST APIs

Cloud, Infra & DX: AWS (Lambda, API Gateway, DynamoDB, S3, Cognito, SNS/SQS, CloudFormation), Firebase, Vercel, Neon, Inngest, Cloudflare R2, Docker, EAS, Fastlane, GitHub Actions, Nx Monorepo, Sentry, PostHog

Testing: Vitest, Jest, Storybook, WebdriverIO, Cypress, Detox, flutter_test, mocktail

EDUCATION

B.Sc. Computer Science — University of Buenos Aires (UBA), 2016–2021 · Senior Technician, Cloud Administration (AWS) — Teclab, 2021–2023